



Startup technical guide

AI agents



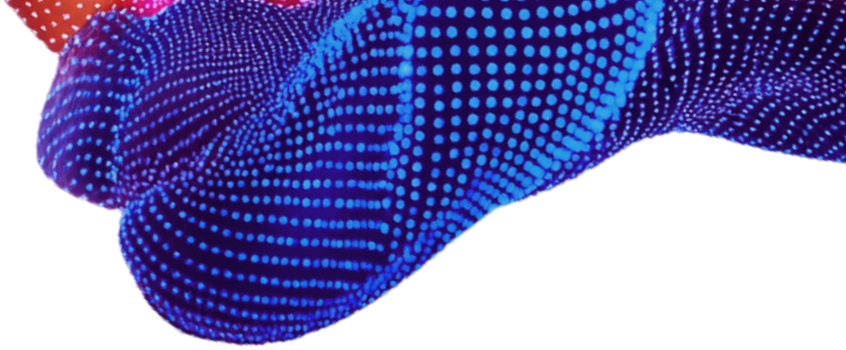


Table of contents

Introduction	01
 Core concepts of AI agents	 02
An overview of Google Cloud's agent ecosystem	04
Key components of every agent	09
The role of grounding in agentic systems	17
Key takeaways	23
 How to build AI agents	 25
A complete toolkit for building AI agents	27
A step-by-step guide: Defining an LLM agent	40
Govern and scale your agent workforce with Google Agentspace	43
Other options for building agents	45
Key takeaways	46
 Ensuring AI agents are reliable and responsible	 48
AgentOps: A framework for production-ready agents	50
Build responsible and secure AI agents with AgentOps	54
Key takeaways	56
 More from Google's full AI stack	 58
 Conclusion	 59
 Resources	 60



Introduction



The development of AI agents represents a paradigm shift in software engineering, enabling startups to automate complex workflows, create novel user experiences, and solve business problems that were previously technically infeasible.

But moving from a promising prototype to a production-ready agent means solving a new set of challenges. How do you manage their non-deterministic behavior? How do you verify their complex reasoning paths? And, crucially, where do you get started?

This technical guide will help answer questions like these. It provides a systematic, operations-driven roadmap for navigating the new landscape, and is geared to help startups and developers who are racing to embrace the potential of agentic systems.

You'll learn the foundational concepts of agentic systems, from their core architectural components to the principles that ensure reliable and responsible operation in production. And you'll learn about the full spectrum of tools that make building and using agents on Google Cloud more efficient, from code-first development with [Agent Development Kit](#) (ADK) and operational automation with the [Agent Starter Pack](#), to no-code agent creation with [Google Agentspace](#).

Whether you're validating an idea, building an MVP, or supporting a product in production, this guide will help across all stages of your project.

How to use this guide

New to AI agents?

Start with [Section 1](#) for the core concepts.

Ready to build?

Jump to [Section 2](#) to create your first agent using ADK.

Agent built?

Dive into [Section 3](#) to make it safe, stable, and scalable.

Want extra support?

Use the [Gemini Kit](#) to prototype faster, and apply to the [Google for Startups Cloud Program](#) to receive expert guidance and up to \$350k USD in cloud credits.

The focus of this guide

The agentic AI ecosystem offers many tools, libraries, and approaches for building cognitive architectures. There are open-source frameworks from Google like [Genkit](#) and [Google Cloud's conversational AI offerings](#), as well as popular open-source libraries like LangChain and CrewAI.

This guide focuses primarily on ADK, sharing concepts and architectural patterns that allow you to build robust, scalable agents on Google Cloud while retaining the ability to integrate other preferred tools and libraries.